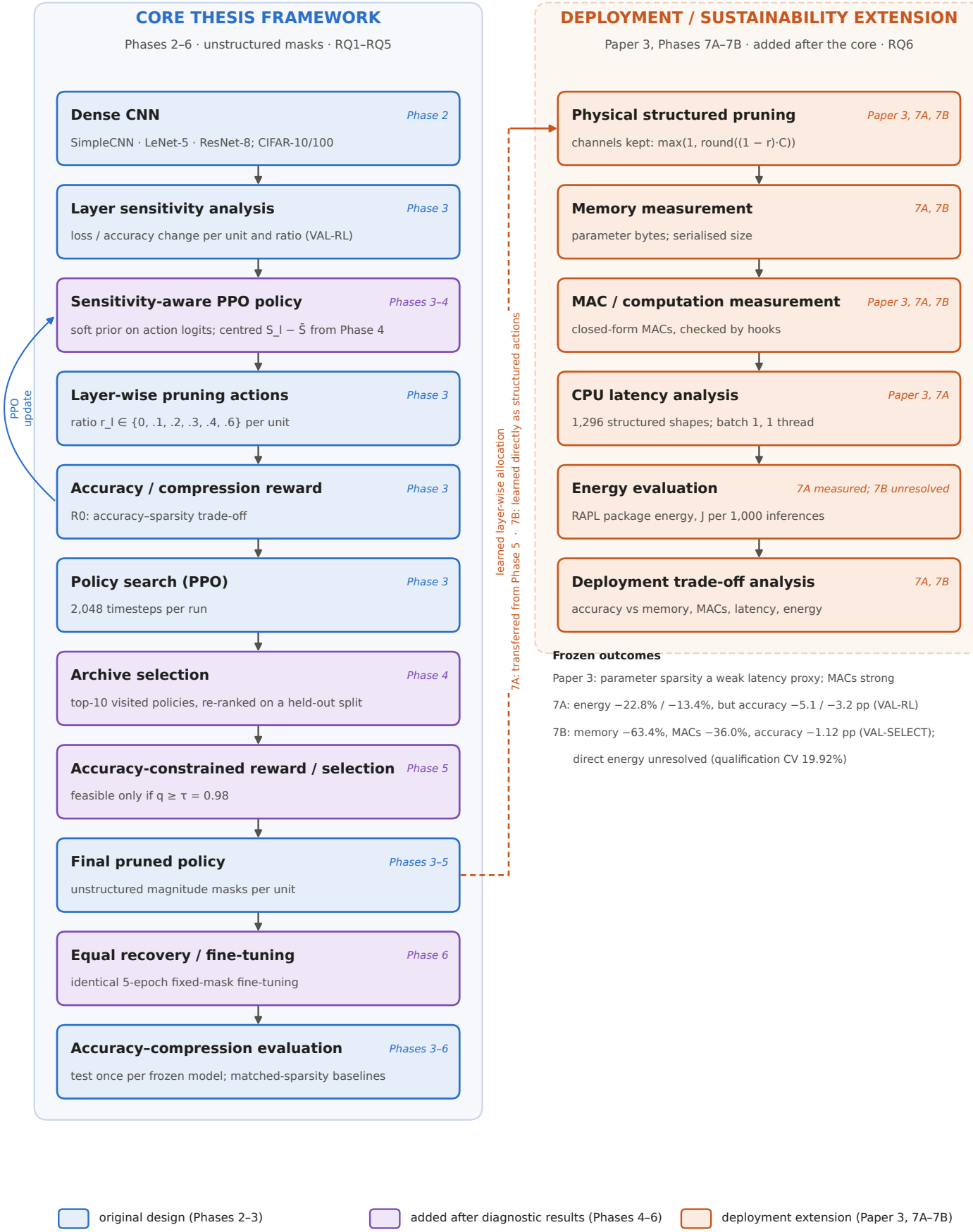

CORE THESIS FRAMEWORK
Phases 2–6 · unstructured masks · RQ1–RQ5
Dense CNN
Phase 2
SimpleCNN · LeNet-5 · ResNet-8; CIFAR-10/100
Layer sensitivity analysis
Phase 3
loss / accuracy change per unit and ratio (VAL-RL)
Sensitivity-aware PPO policy
Phases 3–4
soft prior on action logits; centred S_l − S̄ from Phase 4
Layer-wise pruning actions
Phase 3
ratio r_l ∈ {0, .1, .2, .3, .4, .6} per unit
Accuracy / compression reward
Phase 3
R0: accuracy–sparsity trade-off
PPO update
Policy search (PPO)
Phase 3
2,048 timesteps per run
Archive selection
Phase 4
top-10 visited policies, re-ranked on a held-out split
Accuracy-constrained reward / selection
Phase 5
feasible only if q ≥ τ = 0.98
Final pruned policy
Phases 3–5
unstructured magnitude masks per unit
Equal recovery / fine-tuning
Phase 6
identical 5-epoch fixed-mask fine-tuning
Accuracy–compression evaluation
Phases 3–6
test once per frozen model; matched-sparsity baselines
learned layer-wise allocation
7A: transferred from Phase 5 · 7B: learned directly as structured actions
DEPLOYMENT / SUSTAINABILITY EXTENSION
Paper 3, Phases 7A–7B · added after the core · RQ6
Physical structured pruning
Paper 3, 7A, 7B
channels kept: max(1, round((1 − r)·C))
Memory measurement
7A, 7B
parameter bytes; serialised size
MAC / computation measurement
Paper 3, 7A, 7B
closed-form MACs, checked by hooks
CPU latency analysis
Paper 3, 7A
1,296 structured shapes; batch 1, 1 thread
Energy evaluation
7A measured; 7B unresolved
RAPL package energy, J per 1,000 inferences
Deployment trade-off analysis
7A, 7B
accuracy vs memory, MACs, latency, energy
Frozen outcomes
Paper 3: parameter sparsity a weak latency proxy; MACs strong
7A: energy −22.8% / −13.4%, but accuracy −5.1 / −3.2 pp (VAL-RL)
7B: memory −63.4%, MACs −36.0%, accuracy −1.12 pp (VAL-SELECT);
direct energy unresolved (qualification CV 19.92%)
original design (Phases 2–3)
added after diagnostic results (Phases 4–6)
deployment extension (Paper 3, 7A–7B)